

Bachelor's Thesis

Taxonomy and Analysis of Failure Modes in N-gram Based Generative Retrieval

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Presentation Outline

01

Background

Sparse → Dense → Generative Retrieval · SEAL & MINDER architectures · Datasets

02

Research Questions

RQ1: Taxonomy of failure modes in the literature · RQ2: Empirical analysis of n-gram GR failures

03

RQ1 — Failure Taxonomy

Representation · Training · Inference · Response

04

RQ2 — Empirical Analysis

Non-discriminative DocIDs · Length bias · Metadata dependency · Cultural bias · Query mismatch · Scoring artifacts

05

Discussion & Conclusions

Root cause tracing · Limitations · Future work

Background

From sparse retrieval to generative retrieval

Three Methods of Information Retrieval

1970s–1980s

Sparse Retrieval

e.g. BM25, SPLADE

- Index lookup via term matching
- High-dimensional sparse vectors
- Fast, interpretable, no training needed
- Fails on vocabulary mismatch

2020

Dense Retrieval

e.g. DPR, Contriever

- BERT-based dual encoder
- Query & doc encoded to dense vectors
- Cosine similarity matching
- Requires separate vector index (e.g. FAISS)

2020+

Generative Retrieval

e.g. SEAL, MINDER, DSI

- Corpus encoded into LM parameters
- Autoregressively generates DocIDs
- No external index needed
- Single end-to-end model

Generative Retrieval — How It Works

DocID Types

Numeric

Random integers, hierarchical cluster paths, PQ-codes

Titles

Document title; works well on NQ, fails on MSMARCO

N-grams / Substrings

Any contiguous span of text from the corpus (SEAL, MINDER)

Pseudoqueries

Synthetically generated queries per document (MINDER)

Term Sets

Permutation-invariant bag of terms

Hybrid

Combination of the above

Retrieval Pipeline

1

Corpus Indexing

All documents & passages encoded into LM weights during training

2

Query Encoding

Encoder (BART/T5) embeds the query into a fixed representation

3

DocID Generation

Decoder autoregressively generates DocID tokens via beam search

4

Constrained Decoding

FM-Index or Trie ensures only corpus-valid sequences are generated

5

Passage Ranking

DocIDs are scored to get final ranked passage list

SEAL and MINDER — Analyzed Models

SEAL

Bevilacqua et al.

- DocIDs = any n-gram substring in the corpus
- FM-Index constrains decoding to valid corpus strings
- Scoring: log-odds ratio of $P(\text{ngram}|\text{corpus})$ vs $P(\text{ngram})$ — rewards rare, query-specific n-grams
- Aggregates scores from multiple non-overlapping n-grams per passage
- Limitation: bound to exact word orderings in corpus

MINDER

Li et al.

- Extends SEAL with three identifier views per passage
- View 1: Document title
- View 2: N-gram substrings, same as SEAL (but simpler scoring)
- View 3: Pseudoquery — synthetic query generated for each passage
- Pseudoquery view mitigates lexical gap & compensates for missing metadata

Evaluation Datasets

Natural Questions (NQ)

Google Search queries + English Wikipedia

Clean, unique article titles. Academic benchmark standard. Structured Wikipedia passages. Favours metadata-heavy retrieval strategies.

21M

Passages

6,515

Test queries

MSMARCO

Bing Search queries + Web snippets

Messy, incomplete, non-unique metadata. Reflects real-world search conditions. Harder for models relying on title shortcuts.

8.8M

Passages

6,980

Test queries

Research Questions

RQ1

What failure modes of GR systems have been identified in the literature, and how can they be organized into a coherent taxonomy?

RQ2

Which failure modes appear in n-gram-based retrieval? When and how do they happen?

RQ1 — Failure Mode Taxonomy

Taxonomy Overview — Four Stages

Highlight = Analysis Presented Here

01 Representation

Indistinct DocIDs _{1,2,3,4}

Tokenization failures ₂₉

Static DocID assignment ₁₈

Metadata over-reliance _{14,15}

Lexical mismatch _{19,14}

N-gram contiguity
constraint ₁₄

02 Training

Dynamic corpus /
deletion / forgetting _{5,6,7}

Western-centric bias
_{20,21,22,23}

Zero-shot failure _{19,24}

Distillation errors _{25,26}

Unlabeled documents _{7,27}

Capacity limits _{27,12}

03 Inference

False pruning _{8,9,3}

Latency / compute cost ₁₂

Length bias _{12,14,15}

Generation error
accumulation ₁₂

Retrieval diversity failure
_{2,3}

DocID hallucination _{14,17}

04 Response

LLM hallucination _{10,11}

Context window limits _{4,13}

Lost in the Middle ₁₆

Privacy / PII leakage ₄

Parametric vs. contextual
conflict ₁₃

Citation hallucination ₂₈

RQ1 — Representation Failures

Failure modes rooted in how documents are identified

Indistinct / Non-Unique DocIDs

IVF = "Inverted File Index" or "In Vitro Fertilization" - same identifier but two unrelated documents. In cluster-based DocIDs, "Apple" (company) and "apple" (fruit) can be forced into one cluster.

Tokenization Failures

Rare terms like "Zyloprim" split into "Zy-lo-prim" — model assigns low probability to each meaningless subword. Non-Latin scripts up to 10× more tokens per word than English.

Static DocID Assignment (Non-differentiability)

DocIDs fixed before training begins → indexing module cannot receive gradients. Model mimics a fixed index rather than learning optimal representations.

N-gram Contiguity Constraint

For n-gram-based models, n-grams must be strictly contiguous. Intervening tokens break the span. Passage cutoffs also break span.

Lexical Mismatch

Even with parametric knowledge, query "heart attack" may not trigger DocID "myocardial infarction" if the association was not learned during identifier and model training.

RQ1 — Training Failures

Failure modes emerging from how models are trained and updated

Catastrophic Forgetting

Sequential indexing of new documents causes >25pt accuracy drop on original corpus. ~88% of documents experience at least one forgetting event during training.

Dynamic Corpus / Deletion

New docs unavailable until full retraining. Deleted docs remain in model weights → privacy and compliance risk. Unlike dense retrieval which can re-index instantly.

Western-Centric Bias

BART trained on English Wikipedia + BookCorpus → implicitly defaults to US/European context. T5 similarly. Query 'the capital' may assume USA. NQ and MSMARCO are also Eurocentric.

Zero-Shot Failure

Models trained on NQ perform well on NQ-like queries but degrade sharply on OOD distributions. NCI drops from 71.2 to 27.7 R@100 when evaluated on unseen queries

Capacity Limits at Scale

DSI MRR@10 drops from 80.3 (100k corpus) to 24.2 (8.8M corpus). Adding 8.8M atomic DocIDs adds ~6.8B parameters to output layer alone.

Distillation / Teacher Bias

If cross-encoder teacher is used and misranks a document, the GR student bakes that error into its weights. Teacher errors become quasi-structural biases.

RQ1 — Inference Failures

Failure modes during autoregressive DocID generation

False Pruning (Beam Search)

Beam search discards a DocID prefix if its cumulative probability lags competitors, even if subsequent tokens would have been correct. Once pruned, the path is irrecoverable.

Length Bias

Token probabilities are multiplicative. Each extra token shrinks total sequence probability, drowning out precise identifiers.

Generation Error Accumulation

Error rate rises from ~1% at position 1 to >12% by position 6. Model 'loses its train of thought' generating long DocID sequences.

Retrieval Diversity Failure

Ambiguous queries (e.g. 'What is IVF?') can return topically redundant passages all targeting one interpretation, ignoring others.

DocID Hallucination

Without constraints, model generates DocIDs not present in corpus. Mitigated in practice via FM-Index or Trie. Some methods use intentional pseudoquery hallucination.

Inference Latency

Token-by-token autoregressive generation is FLOP-intensive. Scales poorly with beam size and sequence length vs. simple vector lookups in DR/SR.

RQ1 — Response Generation Failures

Post-retrieval failures in the RAG / answer synthesis stage

Note: these failures belong to the response generation (RAG) stage, not core GR, but included for a complete taxonomy.

LLM Hallucination

Summarizer generates information absent from retrieved passages, relying on parametric knowledge instead.

Context Window Limits

Query requiring 20 sources gets truncated at document 5, so information is discarded.

Lost in the Middle

LLMs over-weigh information at beginning and end of context, ignoring middle documents regardless of relevance.

Parametric vs. Contextual Conflict

Retrieved doc says "Chancellor is Christian Stocker" but LM weights say "Karl Nehammer", model may trust outdated weights.

Citation Hallucination

Model claims "sky is blue [Doc 2]" but the fact is actually from Doc 1. Hurts verifiability and user satisfaction.

Privacy / PII Leakage

Query about address formats returns a real memorized home address from training data → compliance challenge.

RQ2 — Empirical Analysis

N-gram GR failures: when and how do they occur?

RQ2 — Methodology

Output Data

- Top-100 retrieved passages per query
- Matched n-gram identifiers & scores to passages
- Annotated ground-truth passages
- Query & answer strings

Key Signals

- N-gram corpus frequency
- N-gram length (unigrams vs. multi-token)
- Title vs. body vs. pseudoquery score share
- Lexical query-ngram overlap
- Token diversity

Metrics

- Hits@1, @10, @100
- MRR
- nDCG@10, nDCG@100
- Recall@10, @100
- Spearman ρ correlations

6,515

NQ queries

6,980

MSMARCO queries

1M+

Passages analyzed

2

Models compared

RQ2 · Non-Discriminative DocIDs

Model relies on generic n-grams, triangulating instead of identifying

The model generates generic words matching thousands of passages rather than pinpointing the correct one. Forces the model to 'triangulate' from common n-grams instead of directly generating a discriminative identifier.

Deciles created by taking each queries' top result's the top-5 highest scored (= highest weighted) n-grams

28,753

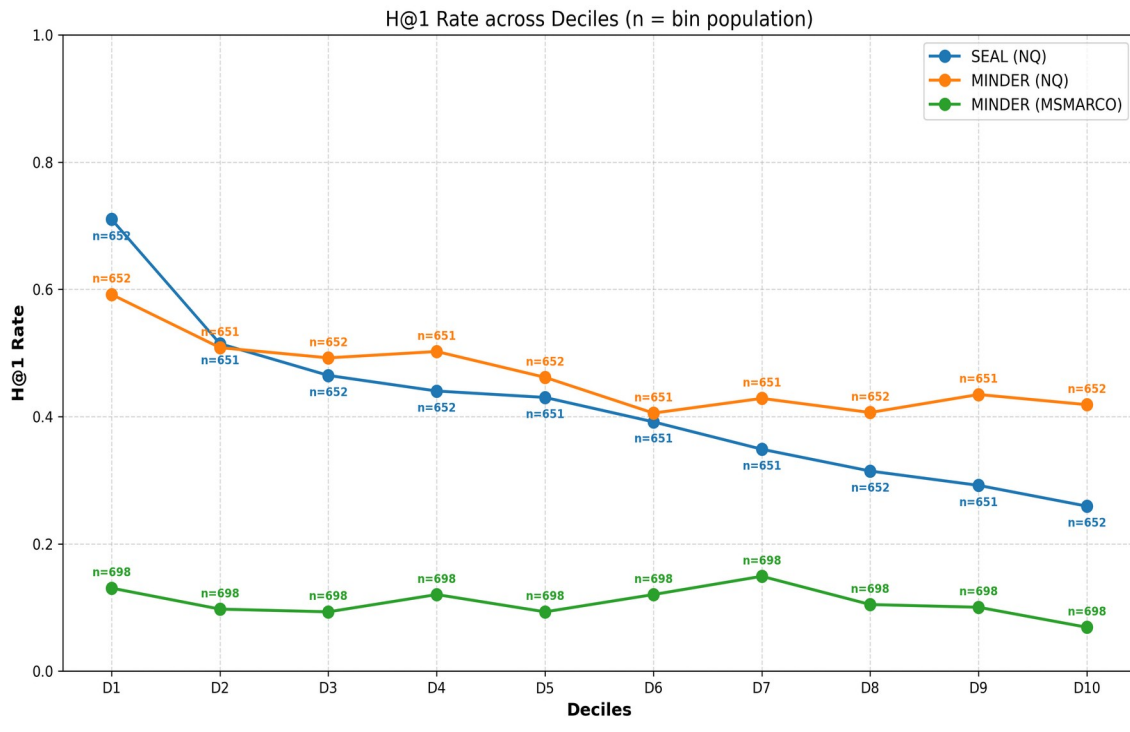
SEAL avg. top-5
n-gram freq. (NQ)

21,310

MINDER avg. top-5
n-gram freq. (NQ)

42,380

MINDER avg. top-5
n-gram freq.
(MSMARCO)



RQ2 · Length Bias

Autoregressive probability penalises long, discriminative n-grams

86.9%

Unigram %
SEAL (NQ)

85.3%

Unigram %
MINDER (NQ)

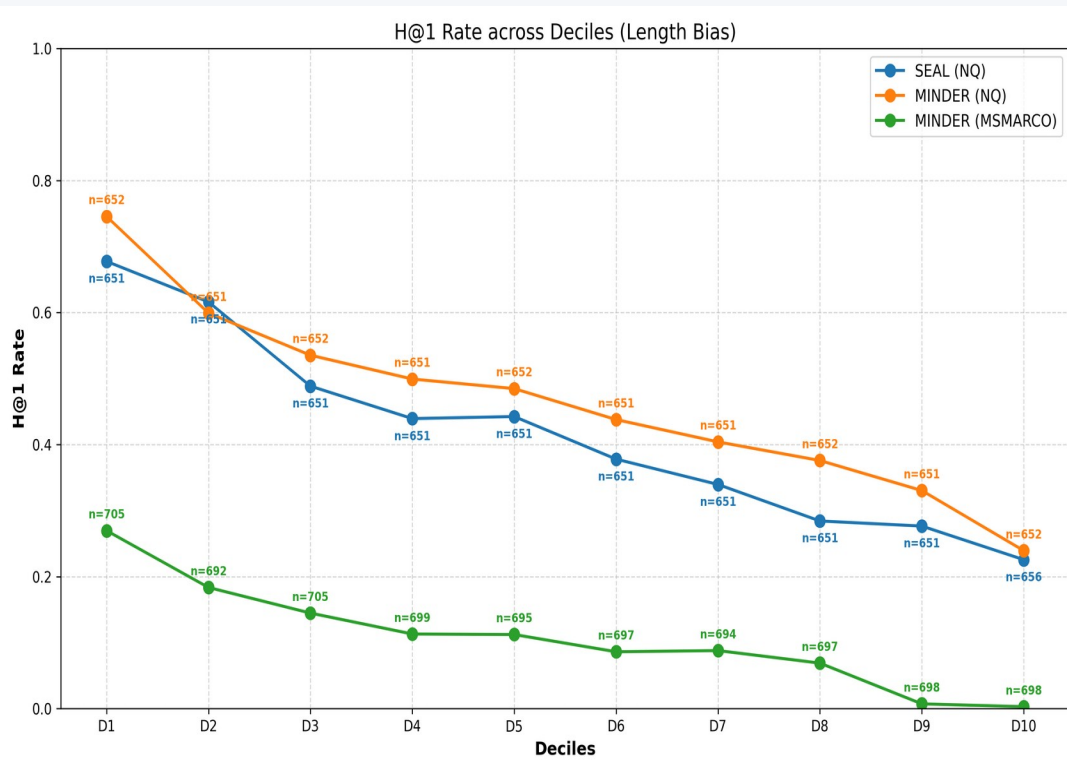
1.28

Avg. n-gram
length (words) —
SEAL

Token probabilities are multiplicative. Each token added to a sequence reduces its total probability score.

The model therefore prefers short, generic n-grams like "War" over precise sequences like "The Napoleonic Wars", as they are statistically safer to generate.

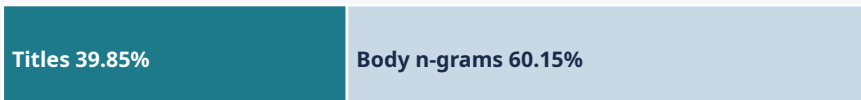
This drowns out the most discriminative identifiers



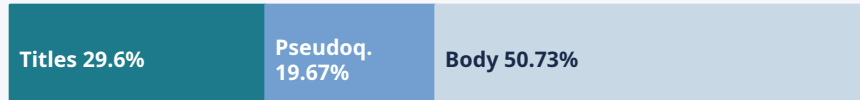
RQ2 · Metadata Dependency

Both SEAL and MINDER over-rely on passage titles as retrieval shortcuts

SEAL (NQ) — % of total score mass



MINDER (NQ) — % of total score mass



Title scoring advantage

When answer appears as title n-gram vs. body n-gram: SEAL scores it 2,866% higher · MINDER: 2,499% higher → title acts as near-perfect signal when present.

Successful retrievals rely less on titles

For correctly retrieved passages, the advantage drops to 1,876% (SEAL) and 1,439% (MINDER) → success correlates with better use of body text.

MSMARCO: performance collapse

SEAL is 30.7% worse Recall@5 than BM25 on MSMARCO. MINDER only 3.1% better. Neither reached 30% Recall@5₁₅ → metadata absence shows shortcut

RQ2 · Cultural Bias

Western-centric training data skews retrieval for non-Western queries

Query: "where did the southern song have their capital?" → Correct answer: Lin'an (Southern Song Dynasty, China, ~1127 AD)

SEAL

- All top-100 results about the southern United States
- 'southern' → US South · 'song' → country music / Dixie · 'capital' → Nashville / Richmond
- Song Dynasty passage not in top-100 at all
- BART trained overwhelmingly on English/Western data

MINDER

- Ground truth appears at rank 4
- All top-100 results relevant to Ancient China
- Pseudoquery: "what was the capital of the song dynasty in north china"
- Success is attributable to the pseudoquery identifier view

Another example: who was the last king of scotland based on (correct answer: Idi Amin, former Ugandan President. SEAL retrieved Scottish heads of state, MINDER succeeded)

Pseudoquery view can partially compensate by generating contextually richer identifiers by counteracting the base LM's pretraining data.

RQ2 · Query–DocID Lexical Mismatch

Coverage $C(Q, G_n)$: fraction of query tokens appearing in generated n -grams

For each query, we took the highest-scored passage's coverage query-docid coverage score, then stratified it into 5 buckets

Spearman ρ : Coverage $C(Q, G_n)$ vs. Hitrate (all $p < 0.001$)

SEAL (NQ)

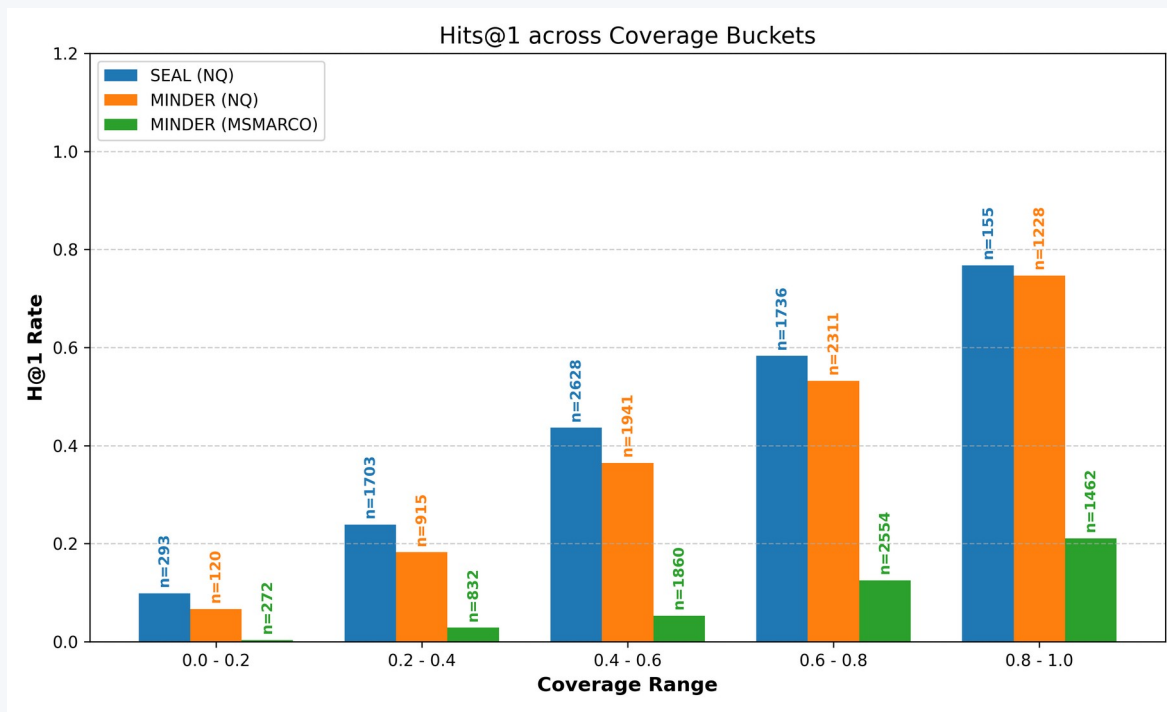
Hits@1: $\rho = 0.323$

MINDER (NQ)

Hits@1: $\rho = 0.380$

MINDER (MSMARCO)

Hits@1: $\rho = 0.223$



RQ2 · Concentration & Token Diversity

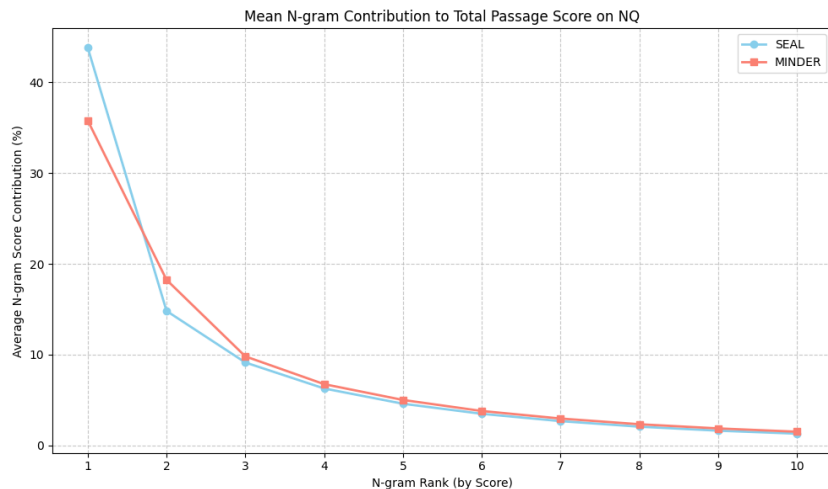
Score Concentration (Single-DocID Overreliance)

Metric: $\text{top } n\text{-gram score} \div \text{sum of all } n\text{-gram scores}$.

Lower ratio (distributed evidence) $\rightarrow$ higher Hits@1.

When one wrong n-gram dominates, remaining low-scoring n-grams cannot always correct the ranking.

MINDER's lower ratio reflects multiview identifiers (titles + pseudoq. + n-grams) distributing score more evenly.



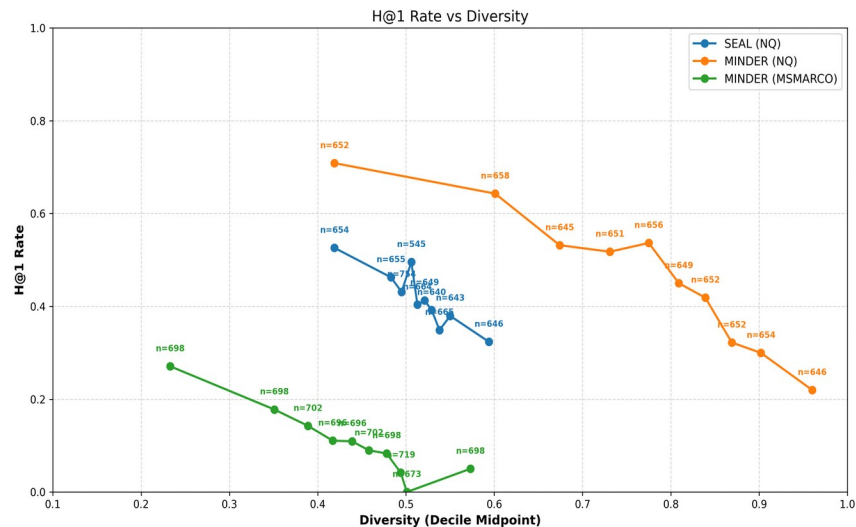
Token Diversity

Metric: $\text{unique tokens} \div \text{total tokens across all generated } n\text{-grams}$.

Result: high overlap = signals model confidence = success.

High redundancy does not lead to lower success rates.

High diversity = loss of focus = failure.



Discussion & Conclusions

Root causes, limitations, and future directions

Limitations & Future Work

Limitations

Empirical analysis limited to SEAL & MINDER on NQ / MSMARCO open-domain QA.

No experiments on other KILT tasks (dialogue generation, fact verification, slot filling, entity linking).

Several GR models analyzed qualitatively only → these require dedicated studies.

Future Work

Multilingual GR: non-English corpora exhibit unique tokenization & cultural bias failure profiles.

Continual learning solutions: dynamic corpora remain an unsolved challenge with massive practical upside.

Alternative architectures: RetNet / Mamba could address latency & scalability without Transformer overhead.

Future Work

Task-specific failure analysis for code retrieval, entity linking, multi-hop QA.

Comprehensive response generation failure taxonomy.

Multimodal GR failure modes (audiovisual retrieval models exhibit unique failure patterns).

Future Work

Skip-gram based DocIDs: relaxes contiguity constraint for storage tradeoff

Permutation-invariant decoding: reduces false pruning sensitivity.

Non-differentiability: Straight-Through Estimator enables end-to-end gradient flow.

Conclusions

RQ1

A taxonomy of GR failure modes across Representation, Training, Inference, and Response stages was developed from the literature.

RQ2-1

N-gram GR models know the semantic neighbourhood of the answer → the dominant failure is discriminative ranking.

RQ2-2

Metadata dependency is a structural training bias: both SEAL and MINDER overweigh titles → a feature on NQ, a failure on MSMARCO.

RQ2-4

MINDER's pseudoquery identifiers help disambiguation and evening out the score distribution, but cannot fully compensate for absent metadata.

Overall

GR promises an elegant architecture, but remains brittle. Solutions to many of its failure modes are not fully achieved. It is promising but not production-ready as of now.

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